

Aircraft Design

Project Description

Introduction

The Aircraft Design Project focuses on the conceptual phase of aircraft design, and it allows the application of the topics and theory presented in the lectures for a real-life, practical open-end engineering problem. The project also aims at developing basic engineering soft skills.

Overall Picture

Aviation is now undergoing a transition, with general aviation moving towards decarbonization, as well as the ever increasing use of unmanned aerial vehicles for diverse missions. **In light of this new emphasis, the course project focuses on the development of a fixed-wing aircraft (no eVTOL or multicopter types) that operates in the fully subsonic ($M < 0.3$) flight regimes.** The type of aircraft to be designed is unspecified, except for aircraft specifically designed for military use (*Read: No military aircraft allowed*), so it is the responsibility of each team to agree on the mission and to determine specifications.

/!\ Both electrical and combustion propulsion systems may be considered (based on the size of the aircraft and the mission.). If a combustion engine is selected, the use of SAF should be considered and the environmental impact of the flight (emission calculations) should be included in the report. It may be assumed that a fully-electrical propulsion will not have any environmental impact during flight /!\

Project Deadlines

The final report is due on November 20th at 17h00. Submissions are only accepted via Canvas. /!\ Periodic milestones will be listed.

Mission Selection and Specifications

- The type of mission and the required specifications for the aircraft are to be chosen by the team.
- The purpose is to allow each group to build “ownership” of their work and to understand what constitutes a “reasonable” specification.

Report Format

- The main body of the report must not exceed 30 pages
- The report must be written in Latex using the provided template from the course website
 - You will need to update the template accordingly.
- The report must include the following sections (as shown in the .tex file):
 - Introduction
 - Initial Sizing
 - Aerodynamic Analysis
 - Performance Analysis
 - Static Stability Analysis
 - Final Configuration
 - References
 - Appendixes

Deliverables to be Included

Deliverables are the requirements for completing the project for the purpose of the course and are a list of items to be included in the final report. The main body of the report should include a discussion of the design process and the main deliverables. It should not be a list of equations and values.

- **Introduction:** the introduction must include the reference literature that supports the choice of mission and final design. For example: if you design a fixed wing UAV for a specific mission, you must include in the introduction a review of existing aircraft (or lack thereof) and the business/societal need for such an aircraft.
- **Initial Sizing:** this section must include a “cloud graph” of key aircraft performance parameters that should be used as starting parameters for the constraint analysis (which is independent of the type of propulsion).
 - Data may need to be “guessed” for smaller aircraft such as fixed-wing UAVs or statistical data may be used from similar larger aircraft as long as it is used in coefficient form.
 - Weight estimation using the “Initial Weight Analysis Method”
 - Maximum Take-Off Weight
 - Empty Weight
 - Payload Weight
- **Aerodynamic Analysis:** this section discusses the aerodynamic design and performance of the aircraft, highlighting how the wing(s) profile(s) and geometries were chosen and the relevant plots and images should be included.
- **Performance Analysis:** this section discusses the performance analysis of the designed aircraft. Depending on the aircraft being designed, statistical data or certification data may be available. For example, when designing an aircraft that fits in an existing category, certification parameters (e.g.: FAR) must be taken into consideration. Otherwise, a reasonable engineering “guestimation” is acceptable. The V-n diagram should be also included.
- **Static Stability Analysis:** this section discusses the longitudinal static stability and control analysis of the aircraft, including control surface sizing. The relevant graphs and images should be included. Only wing(s) + tail(s) need to be considered.
- **Final Configuration:** this section summarizes the design of the aircraft and discusses all design features not previously presented, such as fuselage sizing, landing gear, etc... An image of the final aircraft design should be included in this section.
- **References:** a list of references of books, articles, websites, etc... used in the design process of the aircraft
- **Appendix:**
 - A 3-View drawing of the aircraft from Open VSP
 - A table with the following data:
 - Aircraft Configuration
 - Centre of Gravity (as a function of mean aerodynamic chord)
 - Aerodynamic Centre
 - Wing Geometry
 - Span
 - Planform Area
 - Mean Geometric Chord
 - Aspect and Taper Ratio,
 - LE and c/4 Sweep Angle (if applicable)
 - Dihedral/Anhedral Angle
 - Airfoil Profile(s)

- Twist
- Tail Geometry
 - Span
 - Planform Area
 - Mean Geometric Chord
 - Aspect and Taper Ratio,
 - LE and c/4 Sweep Angle (if applicable)
 - Dihedral/Anhedral Angle
- Aerodynamics
 - Sectional and wing lift curves
 - Lift coefficient vs. angle of attack
 - Maximum lift coefficient
 - Stall angle
 - (Adjusted) Drag Analysis
 - Pressure drag (zero-lift drag and induced drag estimations)
 - Skin friction drag (simplifies calculations for squared wing)
 - Sectional and wing drag polar
 - Efficiency
- Performance
 - Cruise speed
 - Stall speed (with no flaps)
 - Take-Off and Landing run and speed
 - Climb speed and climb rate
 - Maximum glide distance, minimum glide rate, glide angle and glide velocity
 - Load factors (V-n diagram)
 - Maneuvering speed
- Longitudinal Stability and Control
 - Neutral point
 - Static margin
 - Pitching moment coefficient
 - Overall pitching moment
 - Residual pitching moment
 - Longitudinal static derivative
 - Trim angle
 - Tail moment arm
 - Tail efficiency factor
 - Elevator size and control
- Any other graph or image that you think it is necessary to include, with the obligation that it is referred to in the main body of the report.

The template for the final report will be provided after the first progress examination, along with basic writing and editing requirements. **A maximum of 30 pages (excluding appendices) is required for the report.**

Abstract, Table of Contents, Appendices do not count towards the 30 maximum page limit.

Writing and Editing Style

The following writing and editing styles must be followed:

- The report must be written in using the formal impersonal style (e.g.: “Results show...” “The design has been completed...” etc...). No personal form will be accepted (e.g.: “We chose this airfoil because...”).
- All Figures and Tables must be placed either top or bottom of the page nearest to the text to which they refer. This is to keep the format tidy and to optimize space
- All Figures, Tables, and Equations must be clearly labelled as follows:
 - “Fig. No.” if in the text, and “Figure No.” if at the beginning of a sentence (e.g.: Fig. 1 or Figure 1)
 - “Table No.” (e.g.: Table 1)
 - “Eq. No.” if in the text, and “Equation No.” if at the beginning of a sentence (e.g.: Eq. 1 or Equation 1)
- When referring to a section in the text, format as stated for figures and equations (e.g.: Sec. 1 or Section 1)
- Do not use the terms “below,” “above,” “next,” etc... to refer to section, figure, table, equation, etc... BE SPECIFIC (e.g.: do not write “as the figure below shows” but rather “as Fig. 1 shows...”)
- Do not begin a sentence with an acronym or a symbol (e.g.: “The airfoil chosen is NACA0008. C_l is...” but rather “The airfoil chosen is NACA0008. The lift coefficient C_l is...”)
- References must be listed in the sequence in which they appear in the text as you will use the numeric listing. To cite correctly a reference: “The lift coefficient obtained is consistent with available data [4]. Raymer [4] suggests a value of...”

The structure of the report, beyond abstract, is free for the team to decide upon.

Project Organization

OpenVSP (<https://openvsp.org/>) is suggested to be used for the aircraft design process and for the validation of the aerodynamic (and stability) analysis. This is an open-source software developed by NASA for conducting the conceptual design of aircraft. There are several tutorials available to help with the learning of the software (<https://www.youtube.com/@openvspvideos5394>). Open source xflr5 is also accepted.

Project Grading

The project is worth 60 points (40% of the final grade), which are divided as follows:

- 10 points for writing and editing style
- 50 points for overall technical content as requested .
 - If items are missing points will be deducted

/!\ The use of generative AI tools or any other AI tools is strictly forbidden. AI tools may be used for report writing purposes (helping with English grammar and syntax) only and it must be clearly stated in the report in the “Introduction” section.

Course Textbook

1. Gudmunsson, S.: **General Aviation Aircraft Design. Applied Methods and Procedures.** Butterworth-Heinemann, Elsevier, 2nd Edition. (Available online at KTH library). *Available as ebook and hardcopy on Elsevier website. Hardcopy might experience delays in delivery.*

Additional Literature (available at KTH Library and **STRONGLY** recommended to be used)

1. Raymer, Daniel P.: Aircraft Design: A Conceptual Approach, AIAA Educational Series, 2nd Ed., 1992
2. Roskam, Jan: Airplane Design I – VIII, DARCorporation (not updated for electrical aircraft)

3. Gundlach, J.: Designing Unmanned Aircraft Systems, AIAA Educational Series (1st edition: some spelling / editing mistakes).

Teaching Staff

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